BLOOD & GLORY

PHASE 2 DEVELOPMENT PLAN

"Architecting the Arena. Forging the Logic. Unleashing the Chaos."

Architecture Overview

The project utilizes a strict **Entity-Component-System (ECS)** architecture.

Entities	Components	Systems
Objects (Player, Enemy)	Data (Mesh, Light, HP)	Logic (Render, Combat)

Role 1: Graphics Engineer

HIGH DIFFICULTY

Focus: Lighting, Shaders, Materials

- ✓ **Light Component:** Define Type (Dir, Point, Spot), Color, Attenuation & Angles.
  `source/common/components/light.hpp`
- ✓ **Lit Shader:** Blinn-Phong model. Support multiple lights & material maps (Albedo, Specular, Emissive).
  `assets/shaders/lit.frag`
- ✓ **Forward Renderer:** Update to pass light uniforms to shader.

Role 2: World Builder

MEDIUM DIFFICULTY

Focus: Assets, JSON, Level Design

- ✓ **Asset Hunting:** Source OBJ/FBX for Gladiator, Enemy, Arena. Get textures.
- ✓ **Deserialization:** Update JSON loader for `Light` and `Material` props.
- ✓ **Level Design:** Construct `game-scene.jsonc` with Arena, Spawns, Lights (Sun + Torches).

Role 3: Gameplay Programmer

HIGH DIFFICULTY

Focus: Controls, Camera, Physics

- **Orbit Camera:** Implement Zoom/Orbit around target.
  `source/common/systems/camera-controller.cpp`
- **Player Controller:** View-relative WASD movement. Rotation to face movement dir.
- **Collision:** Sphere-Sphere checks. Prevent walking through walls/enemies.

Role 4: Logic & Polish

MEDIUM DIFFICULTY

Focus: AI, Combat, Post-Processing

- **Combat System:** Click to attack. Check distance -> Reduce Health -> Destroy Entity.
- **Enemy AI:** State Machine (IDLE, CHASE, ATTACK).
- **Post-Process:** Implement Vignette or Sepia shader.
  `assets/shaders/postprocess/vignette.frag`

Integration Guide

1. Create new files for systems & components first.
2. Role 1 (GFX) & Role 2 (World) sync up to get a visible scene.
3. Role 3 use "Placeholder" cubes until Role 2 delivers models.
4. Role 4 depends on Role 3's movement code.

GOOD LUCK GLADIATORS!

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